

Performance Test

Test Run ID: 4973f001-aa1f-43e1-b478-8630843cede8

This chapter provides a summary of the test run metrics. The tables contain the aggregated values of the metrics for the entire test run.

Date	04 June 2026	Max VUs Tested	300
Executor	ramping-vus	Tool	k6

1. Executive Summary

The target endpoint for this performance test is the "https://quickpizza.grafana.com" URL. The testing type is a spike test, which involves a sudden increase in load to simulate a large number of concurrent users. The load profile consists of three stages: an initial ramp-up to 300 requests per second (RPS) over 5 seconds, followed by a sustained load of 300 RPS for 15 seconds, and then a gradual decrease to 50 RPS over 10 seconds.

Stable Threshold	Degradation Onset	Critical Point	Peak Throughput
0-479 TPS	N/A	N/A	470-479 TPS

2. Throughput Analysis

TPS Interval Analysis			
TPS Bucket	Total Requests	Failed Requests	Error Rate %
0-9 TPS	0	0	0.00%
100-109 TPS	218	0	0.00%
130-139 TPS	264	0	0.00%
190-199 TPS	397	0	0.00%
210-219 TPS	436	0	0.00%
220-229 TPS	446	0	0.00%
230-239 TPS	460	0	0.00%
240-249 TPS	972	0	0.00%
470-479 TPS	1,912	0	0.00%

3. Performance Summary

Trends							
metric	avg	max	med	min	p90	p95	p99
http_req_duration	257.82ms	1197.15ms	254.67ms	244.55ms	264.19ms	268.64ms	-
http_req_waiting	256.98ms	560.70ms	254.40ms	244.25ms	263.92ms	268.03ms	-
http_req_connecting	13.59ms	346.46ms	0.00ms	0.00ms	0.00ms	247.08ms	-
iteration_duration	1287.30ms	2198.29ms	1256.64ms	1245.16ms	1271.05ms	1752.73ms	-

Counters

metric	count	rate
data_received	19.39 MB	636.83 kB/s
data_sent	803.09 kB	25.76 kB/s
http_reqs	5,632	180.66/s
iterations	5,632	180.66/s

Rates

metric	rate
checks_succeeded	100.00% (5632/5632)
checks_failed	0.00% (0/5632)
http_req_failed	0.00% (0/5632)

Gauges

metric	value
vus	17 (min: 17, max: 300)
vus_max	300 (min: 300, max: 300)

Checks

Check Name	Passed	Failed	Success Rate	Status
status is 200	5632	0	100.00%	Pass

4. Advanced Analysis

4.1 Error Analysis

- No error or warning logs found in the test execution.
- No metrics abnormalities, error rate spikes, or failed checks detected.

4.2 Observation

The performance test results indicate a successful execution with no errors or warnings. However, upon closer inspection of the metrics, we notice that the average duration of HTTP requests (`http_req_duration`) is relatively high, with an average value of 257.82ms. This could be due to various factors such as:

- High latency in the backend services or databases.
- Inefficient routing or load balancing.
- Insufficient caching or content delivery network (CDN) configuration.
- Overly complex or resource-intensive business logic.

Additionally, the `http_req_connecting` and `http_req_tls_handshaking` metrics show significant values, indicating that a substantial portion of the request duration is spent on establishing connections and performing TLS handshakes. This could be optimized by:

- Implementing connection pooling or reuse.
- Enabling TLS session resumption.
- Optimizing the TLS handshake process.

4.3 Final Recommendations

1. **Advanced Connection Management:** Implement connection pooling or reuse to reduce the overhead of establishing new connections for each request.
2. **Session Persistence sticky affinity/Redis cache:** Implement session persistence using sticky affinity or Redis caching to reduce the load on backend services and improve response times.
3. **Gateway resilience Istio outlier circuit breaker:** Implement Istio's outlier detection and circuit breaker features to detect and prevent cascading failures in the gateway.
4. **Observability tracing and standardized log levels:** Implement observability tracing and standardized log levels to better understand the performance bottlenecks and identify areas for optimization.
5. **Database read/write splitting and PgBouncer connection pooling:** Implement database read/write splitting and PgBouncer connection pooling to reduce the load on databases and improve response times.

4.4 Conclusion

The performance test results indicate a successful execution with no errors or warnings. However, the high average duration of HTTP requests and significant values in the `http_req_connecting` and `http_req_tls_handshaking` metrics suggest areas for optimization. Implementing the recommended changes will help improve the overall performance and responsiveness of the system.